

Let G be res. discrete



Let $(Q_i)_{i \in \mathbb{N}}$ be compactly generated ^{open} and s.t. $\downarrow$

$$G = \bigcup_{i \geq 1} Q_i \quad \left(\mathcal{G}(G) = \sup_i \{ \mathcal{G}(Q_i) \} + 1 \right) \quad \text{if } G = Q_i \text{ for some } i \Rightarrow$$

$$\mathcal{G}(\text{Res}(Q_i))$$

Let $g \in Q_i \Rightarrow \exists Q_g \not\subseteq G$ s.t. $g \notin G$

$\Rightarrow Q_g \cap Q_i \not\subseteq Q_i$ with $g \notin Q_g \cap Q_i$

$\Rightarrow Q_i$ is res. discrete

$\Rightarrow \text{Res}(Q_i) = 1 \Rightarrow \mathcal{G}(\text{Res}(Q_i)) = 1 \Rightarrow \mathcal{G}(G) \leq 2$

7.8

~~7.8~~

$H \leq G$, H elementary, $N_G(H)$ open. Show:

$\exists Q \leq G$ s.t. $H \trianglelefteq Q$ and $\mathcal{G}(Q) = \mathcal{G}(H)$

Let $Q = N_G(H)$

$\Rightarrow H \trianglelefteq Q$, is $H \trianglelefteq_{cc} Q$?

Let $U \leq N_G(H)$, $U \in \mathcal{U}(G)$

Then $UH = HU$ and $H \trianglelefteq HU$

and $HU/UH \cong \overline{U/U \cap H} \Rightarrow H$ is cocompact

and $\mathcal{G}(H) = \mathcal{G}(UH)$

~~7.11~~ 7.11

$$\bigoplus_{i \in \mathbb{N}} (G_i, U_i) \in \bullet \Sigma \quad \left(= \bullet \bigcup_{i \in \mathbb{N}} S_i \text{ where } \right.$$

$$S_0 = \prod_{i \in \mathbb{N}} U_{a_i}$$

$$S_{n+1} = G_{a_0} \times \dots \times G_{a_n} \times \prod_{i > n+1} U_{a_i} \bigg)$$

For each $n \in \mathbb{N}$ let

$$(\mathcal{Q}_i^{(n)})_{i \in \mathbb{N}} \text{ be comp gen. gr. subgroups s.t. } \bigcup_{i \in \mathbb{N}} \mathcal{Q}_i^{(n)} = G_n$$

Define

$$\mathcal{Q}_i =$$

$$\mathcal{Q}_0 = \prod_{i \in \mathbb{N}} U_{a_i}$$

$$\mathcal{Q}_{i+1} = \mathcal{Q}_i \times \mathcal{Q}_{i+1} \times \dots$$

$$\begin{array}{c} \nearrow \text{all comp. gen.} \\ \mathcal{Q}_{i+1}^{(0)} \times \mathcal{Q}_{i+1}^{(1)} \times \dots \times \mathcal{Q}_{i+1}^{(n)} \times \prod_{k > i+1} U_{a_k} \end{array}$$

$$\Rightarrow \bigcup_{i \in \mathbb{N}} \mathcal{Q}_i \text{ is s.t. } S_i \subseteq \bigcup_{i \in \mathbb{N}} \mathcal{Q}_i, \forall i \in \mathbb{N} \Rightarrow \bigoplus_{i \in \mathbb{N}} (G_i, U_i) = \bigcup_{i \in \mathbb{N}} \mathcal{Q}_i$$

because $\mathcal{Q}_i$ is product of compact groups and ^{finitely many} elementary groups it is elementary $\Rightarrow G$ is elementary

(if V_1, V_2 elementary

$$\{1\} \rightarrow V_1 \rightarrow V_1 \times V_2 \rightarrow V_2 \rightarrow \{1\} \Rightarrow V_1, V_2 \text{ elementary})$$

7.12

If $|G| = n \Rightarrow$ Trivial

Let $|G| \geq 11$

If G is discrete $\Rightarrow$ Trivial

If G is not trivial $\Rightarrow \exists U \in \mathcal{U}(G)$ s.t. $U \neq \{1\}$

$\exists K \subseteq G, K$ comp, $\langle K \rangle = G$

Let ~~some~~ $V \in \mathcal{U}(G), V \neq 1$

$$\Rightarrow \bigcup_{K \in \mathcal{K}} K V \supseteq K$$

$\Rightarrow \exists K_1, \dots, K_m$ s.t.

$$\bigcup_{i=1}^m K_i V \supseteq K$$

$$u \in \bigcap_{i=1}^m K_i V K_i^{-1}$$

$$\Rightarrow K_{i_1} u K_{i_1}^{-1} \in \bigcap_{i=1}^m K_i V K_i^{-1}$$

$$\Rightarrow U' = \bigcap_{i=1}^m K_i V K_i^{-1} \trianglelefteq G ?$$

Let $g \in G, g = K_{i_1} K_{i_2} \dots K_{i_n}$

$$K_{i_1} K_{i_2} \dots K_{i_n} K_i V K_i^{-1} K_{i_1}^{-1} K_{i_2}^{-1} \dots K_{i_n}^{-1}$$

$$\Rightarrow U' \trianglelefteq G$$

7.13 Want $G/\text{Res}(G)$ is elementary

~~Let $(A_i)_{i \in \mathbb{N}}$ be s.t.~~

~~$\bigcup_{i \geq 1} A_i = G$, $A_i \cap A_j = \emptyset$ (0) can be 0 after some point?~~

~~Let $\mathcal{U}$ compact/open~~ Let $(R_i)_{i \in \mathbb{N}}$ s.t. They are comp.

generated and $G = \bigcup_{i \in \mathbb{N}} R_i$, as in 7.12 get $U \in \mathcal{U}(G)$

s.t. U is compact, open, normal

$$\Rightarrow \bigcup_{i \in \mathbb{N}} U R_i = G$$

$$\text{in a similar way } \bigcup_{i \in \mathbb{N}} U R_i / \text{Res}(G) = G / \text{Res}(G)$$

want: $\forall g$

Let $g \in G / \text{Res}(G) \Rightarrow \nexists \emptyset \leq G$ s.t. $g \notin \emptyset$

$\text{Res}(G)$ elem.

$G / \text{Res}(G)$ elem $\Rightarrow$

$$\{1\} \rightarrow \text{Res}(G) \rightarrow G \xrightarrow{\text{elem.}} G / \text{Res}(G) \rightarrow \{1\}$$

7.19

G loc. ell. $\Rightarrow \forall g_1, \dots, g_n \in G, \overline{\langle g_1, \dots, g_n \rangle}$ compact

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~~$G = \bigcup_{n \in \mathbb{N}} \overline{\langle g_1, \dots, g_n \rangle}$~~

Let $\{g_i\}_{i \in \mathbb{N}}$ dense set in $G, U \in \mathcal{U}(G)$

$\Rightarrow \langle g_1, \dots, g_n \rangle U$ is compact $\forall n \in \mathbb{N}$

$\Rightarrow G = \bigcup_{i \in \mathbb{N}} \underbrace{\langle g_1, \dots, g_i \rangle U}_{\text{all elementary}}$

7.20

Let G be solvable

$$G \triangleright G^{(1)} \triangleright G^{(2)} \triangleright \dots \triangleright G^{(n)} = \{1\}$$

Let G be abelian ($\Rightarrow G^{(1)} = \{1\}$)

~~$\bullet$ If $\mathcal{G}(G) = 2(2)$~~

~~We know: $\forall U \in \mathcal{U}(G), U \trianglelefteq G$~~

~~Let $\{U_i\}_{i \in \mathbb{N}}$ s.t. $G = \bigcup_{i \geq 1} U_i$ and, $\rightarrow$ they all are compact and is $\mathbb{R}$~~

~~because $N_G(U_i) = U_i$, we have $\mathcal{G}(G) \leq \mathcal{G}(U_i) + 1$~~

observe that $\forall U \in \mathcal{U}(G), U \trianglelefteq G \Rightarrow$ we have a basis of $\mathfrak{m}_G$ of $1 \leq i \leq l$. it is invariant under conjugation

$\Rightarrow G$ is SIN $\Rightarrow G$ is residually discrete $\Rightarrow \mathcal{G}(G) \leq 2$

suppose for G s.t. length is n we have $\mathcal{G}(G) \leq 2n+1$
 let G s.t. length is $n+1 \Rightarrow G^{(1)}$ has length n and

$$\{1\} \rightarrow G^{(1)} \rightarrow G \rightarrow \overbrace{G/G^{(1)}}^{\text{ab}} \rightarrow \{1\}$$

$$\Rightarrow \mathcal{G}(G) \leq \mathcal{G}(G^{(1)}) + \mathcal{G}(G/G^{(1)}) = 2n+3 = 2(n+1) + 1$$

Q

ob: $V \leq g \geq V$ is compact / open

$$U \langle g \rangle U = \bigcup_{n \in \mathbb{Z}} U g^n U \Rightarrow \exists k_1, \dots, k_n \in \mathbb{Z} \text{ s.t.}$$

$$\bigcup_{k_i} U_{g^{k_i}} U = U \langle g \rangle U$$

Let $U_g U \neq U_h U \Rightarrow \exists u_1, g u_2$ s.t. $u_1, g u_2 \notin U_h U$

Let $u_1, u_2 \in U \cap U \Rightarrow 1$ because $UU = U$

we have $g \in U_h V$ and $u_3 \wedge u_4 \in U_h V \forall u_3, u_4 \in V$

$\Rightarrow$ if $U_g V \cap U_h V \neq \emptyset$ we have $U_g V = U_h V$

$$g P_1(6) \subseteq P_1(6) \Rightarrow \underline{g^{-1} P_1(6) \subseteq P_1(6)}?$$

$$h \in P_1(G) \Rightarrow \langle \overline{g^{-1}h} \rangle \subseteq \bigcup_{n \in \mathbb{Z}} (\langle \overline{g} \rangle \langle h \rangle)^n \quad \text{not good}$$

Let V be a T -bimodule for U

$$hU = Uh \Rightarrow \langle h \rangle U = U \langle h \rangle \quad \text{compact / open / subgroup}$$

$\langle g^{-1}h \rangle$ AA AAAA AAAA AA A A A A A A A A A